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22 July 1974
SUPERSEDING
MIL-L-16879B(SHIPS)
11 January 1963
(See 6.4)

MILITARY SPECIFICATION

LANTERN, ELECTRIC, DRY BATTERY, HAND CARRYING AND HEAD ATTACHING

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements for a dry battery, electric lantern for hand carrying and head attaching.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
L-P-1183 - Plastic Molding Material, Acrylonitrile - Butadiene - Styrene (ABS), Rigid.

MILITARY
MIL-B-18 - Batteries, Dry.
MIL-B-18/9 - Battery, Dry, BA-30.
MIL-E-17555 - Electronic And Electrical Equipment, Accessories, And Repair Parts: Packaging And Packing of.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Insection by Attributes.
MS15587 - Lamp, Incandescent, G-4-1/2 Miniature Screw.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical group and using Federal agencies.)

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production 2 samples shall be examined and tested as specified in 4.2 (see 6.2).

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3.2 General requirements.

3.2.1 Material and design. The material and design of this lantern shall be in accordance with this specification and in general conformance with figure 1. Exposed metal parts shall be protected against corrosion.

3.2.2 Construction. The complete lantern shall be as simple and as rugged as practicable. Assembly shall consist of a battery case, lamp reflecting unit and connecting cable. Assembly shall be furnished complete with lamps but without batteries. It shall be of the nonshort-circuiting type, that is, the battery circuit cannot be closed by metallic contact outside the case. Insulation shall be a thermosetting plastic material (paper base).

3.3 Detail requirements.

3.3.1 Battery case. Battery case shall be made of either sheet metal or a plastic compound. Plastic shall conform to L-P-1183, ABS material. Wall thickness shall be not less than 0.025 inch, if metal, or 0.156 inch if plastic. The case shall be designed to receive 4 standard type BA-30 cells conforming to MIL-B-18 and MIL-B-18/9.

3.3.1.1 Belt clip. A molded plastic or metal belt clip shall be fastened to the side of the case to permit attaching to belt for carrying.

3.3.1.2 Latching device. The case shall be provided with a latching device which will positively hold the cover closed.

3.3.1.3 Instruction label. An instruction label shall be fastened inside the case which will clearly indicate the proper position of the battery cells and type batteries to use.

3.3.2 Lamp reflecting unit. Lamp reflecting unit shall consist of the lens, lens retaining ring, reflector, lamps and holders, lamp housing, bracket, resilient pad, head band and essential hardware. Lamp housing shall be mounted in the bracket s that vertical adjustment of the light beam may be made. Bracket shall be cushioned with a resilient pad. An adjustable elastic head band (1 inch wide maximum) shall be fastened to one side of the bracket with a clip for attaching to the other side of the bracket. Head band shall be equipped with a clip for holding the cable in such a position as not to obstruct the wearer's vision. Unless otherwise specified in the contract or order, the lens retaining ring, lamp housing and bracket shall be made of sheet aluminum or plastic conforming to L-P-1183.

3.3.2.1 Lamp housing. Lamp housing shall contain the lampholder and the clip for the spare lamp. Size shall be adequate to provide space for cable entrance, attachment of leads to lampholder, space for spare lamp, and depth for reflector adjustment.

3.3.2.2 Lens and retaining ring. Lens retaining ring in which the lens and reflector are mounted shall be attached to the lamp housing by the threads specified in 3.3.2.3. Lens shall be made of a clear acrylic plastic, 3/32 inch thick, free of defects or blemishes which would affect the light pattern. Lens and the reflector shall be secured in the lens retaining ring in such a way that loss will not be encountered when unit is relamped.

3.3.2.3 Threads. Threads shall be at least 10 to the inch. If the retaining ring is metal, threads shall be rolled in accordance with Handbook H28. At assembly, threads shall be lightly coated with silicon grease.

3.3.2.4 Reflector. Metal reflector shall have a highly polished conical or parabolic surface and lacquered. The reflecting surface shall be smooth and free from bubbles or pits. The beam of light shall be evenly diffused and practically free of filament images.

3.3.2.5 Lampholder. Lampholder shall be a commercial type designed to receive the lamp specified in 3.4 and shall be capable of being securely attached to the lamp housing.

3.4 Lamps. Lamps used with this lantern shall be industry number 27 conforming to MS15587. The spare lamp shall be installed in the clip provided.

3.5 Focus. Lamp in its holder shall be located in the axis of the reflecting surface such that the filament of the lamp is at the focal point of the reflector. The light shall be focused by adjusting the lens retaining ring.

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3.5.1 With the lamp positioned as required in 3.5 and the lantern located 5 feet from a screen, the plane of which is perpendicular to the optical axis of the lantern, shall project a concentrated beam of light not less than 5 nor more than 11 inches in diameter.

3.6 Cable. Cable shall be close coiled retractable extendable type SJO-18/2, Underwriters' Laboratory (UL) approved. Length of the cable, when fully retracted, shall be 15 inches. Cable shall be suitably anchored at the battery case and the lamp reflecting unit to prevent "pull out" under normal handling.

3.6.1 Electrical connections. Wire leads connected to the terminals of the lampholder and at the contact strips in the battery case shall be attached as to insure electrical continuity when a pull of 10 pounds is exerted on the conductor.

3.7 Switch. The positive acting switch shall be mounted on either the top of the battery case or the lamp housing. It shall provide an "OFF" and "ON" position. The spring operating parts shall be of phosphor-bronze or spring brass.

3.8 Weight and size. Total weight of the lantern complete less batteries, shall not exceed 1 pound. The dimensions of the battery case, not including belt clip and latching device, shall not exceed 3 x 6-5/8 x 1-5/8 inches. Dimensions of the lamp reflecting unit, not including bracket, shall not exceed 2-3/4 inches diameter by 2 inches depth.

3.9 Marking. Each battery case shall be permanently and distinctly stamped or marked with the manufacturer's name or trade mark.

3.10 Workmanship. Workmanship shall be first class in every respect. Parts shall be free from blemishes, deformations, and other defects which would affect the appearance or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. First article inspection shall consist of the examination and tests specified in 4.5 and 4.6.

4.2.1 Test reports. The supplier shall prepare a first article test report in accordance with the data ordering documents included in the contract or order (see 6.1.2).

4.3 Sampling for quality conformance inspection.

4.3.1 Lot. For purposes of sampling, a lot shall consist of all lanterns offered for delivery at one time.

4.3.2 Sampling for examination. Sample lanterns shall be selected at random from each lot in accordance with MIL-STD-105 at inspection level I for the examination of 4.5. The acceptable quality level (AQL) shall be 2.5.

4.3.3 Sampling for tests. Sample lanterns shall be selected at random from each lot in accordance with MIL-STD-105 at inspection level S-2 for the test specified in 4.6.1.

4.4 Quality conformance inspection. Sample lanterns shall be examined and tested as specified in 4.5 and 4.6.2.

4.5 Examination. Each sample lantern shall be subjected to a surface examination to ascertain that shape, material, dimensions, marking, weight, and workmanship are in conformance with this specification. The fit of parts shall be observed with particular reference to the interchangeability of standard replacement parts. Defects are classified as shown in table I.

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Table I - Classification of defects.

Categories	Defects
Critical:	None defined.
Major:	
101	Materials defective or not as specified.
102	Incomplete, parts missing.
103	Parts bent, dented, cracked, split, broken or marred.
104	Screw threads crossed, not smooth close fitting.
105	Riveting not neatly and thoroughly done, loose, or rivets missing.
106	Battery case defective.
107	Belt clip for battery case missing, improperly located, or not securely attached.
108	Battery case cover latching device fails to hold cover in place.
109	Lamp housing defective or not as specified.
110	Lamp housing bracket not securely attached, or not as specified.
111	Lamp housing bracket not vertically adjustable, or not cushioned as specified.
112	Elastic head band missing, defective, or not as specified.
113	Lamp missing, broken, or not size specified.
114	Spare lamp not mounted in clip behind reflector.
115	Lamp holder defective.
116	Reflector not highly polished and lacquered and corrosion-resistant.
117	Lamp not properly located with respect to reflector.
118	Lens not free from bubbles, crackes, striae, checks, wrinkles, or blemishes.
119	Lens and reflector not secured to the lens ring as specified.
120	Lens ring not adjustable (for focusing).
121	Switching mechanism loose or too tight, not mounted as specified.
122	Switching mechanism spring operating parts not phosphor-bronze or spring brass as specified.
123	Cable defective, not suitably anchored at entrance to battery case and lamp housing, not of sufficient length, or not as specified.
124	Head band not provided with a clip for holding the cable.
125	Thickness of case material (minimum average) less than specified.
126	Inside dimensions of case not sufficient to allow for mounting four standard cells.
127	Lantern not nonshort-circuiting, insulation not as specified.
128	Finish defective or not as specified, (if metal), not coated to prevent corrosion.
129	Total weight, including battery case, and lamp reflecting unit, exceeds specified weight.
Minor:	
201	Marking, manufacturer's name or trademark missing, illegible, incorrect, or not permanently marked.
202	Instruction card missing, illegible or not fastened to the inside of the case.

4.6 Test procedures.

4.6.1 Operation. Sample lanterns shall be fitted with batteries and operated in the dark to observe the concentration and even distribution of light throughout the beam. Lanterns shall be subjected to a photometric test to determine compliance with 3.5.1.

4.6.2 Test to determine nonshort-circuiting of case. This test shall consist of making a metallic contact between the case and the lamp housing and between the case and the switch by means of a wire or some other electrical conductor.

4.6.3 Test of electrical connections. A pull of 10 pounds shall be applied, and electrical continuity attested to, in accordance with 3.6.1.

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4.7 Possible test failures. Possible test failures are defined as follows:

Operation test:
Light beam not evenly distributed, not of sufficient concentration, or not reasonably free of filament images when lantern is operated in the dark.
Light beam now in compliance with 3.5.1.
Test to determine nonshort-circuiting of case:
Circuit can be closed by making exterior metallic contact as specified.
Test of electrical connections:
Electrical continuity is lost during 10 pound pull.

4.8 The examination and tests of 4.5 and 4.6 may not be required under the following conditions:

(a) Equipment offered for delivery by contractors who have previously furnished (on a prime contract) material of identical design. The contractor shall be required to furnish proof (test data) to NAVSEC via the cognizant Defense Contract Administration Service (DCAS) that the product has previously passed the tests.
(b) Items such as cable wire, connectors, clips, tape, and so forth. The contractor shall furnish data to the DCAS that the material is:
(1) UL approved.
(2) Conforms to known commercial standards.
(3) Complies with specification requirements.

4.9 Inspection of preparation for delivery. The packaging, packing and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3.)

5.1 Preservation and packaging. Preservation and packaging shall be level A or C in accordance with MIL-E-17555 as specified (see 6.1.1).

5.2 Packing. Packing shall be level A, B or C in accordance with MIL-E-17555 as specified (see 6.1.1).

5.3 Marking. Marking shall be in accordance with MIL-E-17555.

6. NOTES

6.1 Ordering data.

6.1.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Levels of preservation and packaging, and levels of packing (see 5.1 and 5.2).

6.1.2 Contract data requirements. When this specification is used in a procurement invoking the provisions of Armed Service Procurement Regulations (ASPR) paragraph 7-104.9(n) and which incorporates a DD Form 1423 Contract Data Requirements List (CDRL), the following data requirement should be specified for delivery on the cited Data Item Description (DID) and delivered in accordance with such CDRL. When the ASPR provision are not invoked, the following data should be specified for delivery in accordance with the contract or order.

Spec paragraph	Data requirement	Service	Applicable DID	Options
(a) 4.2.1	Test report first article	SH	UDI-T-23450	-------

(Copies of DID's required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

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6.2 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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Figure 1 - View of electric, dry battery, hand carrying and head attaching lantern.

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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